Day 4.

1. open your own laptop or desktop & Identify the GPU model installed. write down the GPU's brand, model number and whether it is integrated or dedicated.

→ GPU's brand name :- Intel (R) Arc (TM) Graphics

model number :- Laptop - HU9FOIN9

My laptop lenovo Ideapad slim 5 14IMH9 is integrated

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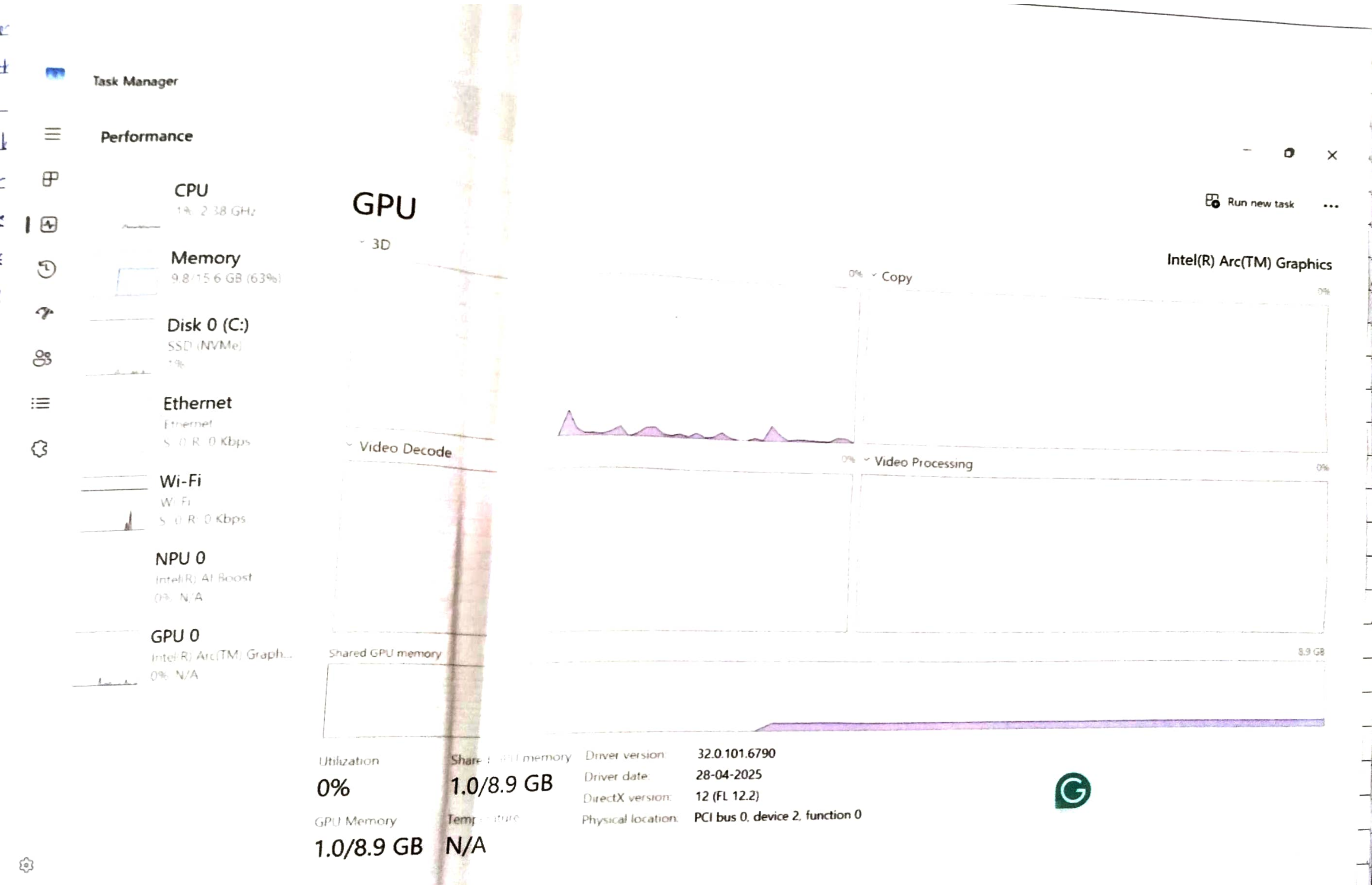
2. Take a clear photo or screenshot of the cooling system. inside your pc or from a pc Hardware simulator. Label the main cooling components and brefly describe their function.


Task Manager
Performance
CPU
Memory
9.8/15.6 GB (63%)
Disk 0 (C:)
SSD (NVMe)
Ethernet
Wi-Fi
NPU 0
Intel(R) AI Boost
0% N/A
GPU 0
Intel(R) Arc(TM) Graph...
0% N/A
GPU
Run new task
Intel(R) Arc(TM) Graphics
3D
Copy
Video Decode
Video Processing
Shared GPU memory
8.9 GB
Utilization
0%
GPU Memory
1.0/8.9 GB
1.0/8.9 GB
N/A
Driver version
32.0.101.6790
Driver date
28-04-2025
DirectX version
12 (FL 12.2)
Physical location
PCI bus 0, device 2, function 0

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3. create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPU's including factors like cost, efficiency and maintenance.

→ Air cooling & liquid cooling are two different methods used to keep a graphics processing unit (GPU) from overheating.

Factor	Air cooling	Liquid cooling
cost	It is cheaper. comes pre installed on most GPUs or costs significantly less as an upgrade.	It is More Expensive. Requires additional, costly parts like pumps, tubes and radiators.
Efficiency & Temperature	It is good for standard use uses fans to blow air over metal fins. works great for daily gaming, but runs warmer under heavy loads. (around 70-80'c)	~~SNPP~~ Superior cooling : uses liquid to absorb heat directly from the GPU and move it away to a radiator. keeps temperatures much lower. (around 55-65' c)
maintenance & longetivity	very Easy : simple to maintain. Just blow out dust with compressed air every few months. Fans rarely fail, and there is zero risk of fluid leaks	Higher unkeep : Requires checking for pump failures or fluid leaks custom keep setups need regular liquid refilling and tube cleaning.
Noise level	It is louder.	It is Quieter.

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4. Imagine you are building a gaming pc for streaming IPL matches and playing graphics heavy games. Choose between air cooling and liquid cooling for your CPU, and justify your choice in 3-4 sentences based on research.

→ For a streaming and heavy gaming PC, I would choose liquid cooling for the CPU. Streaming IPL matches while playing high-end games places a heavy, continuous workload on both the graphics card and processor, producing a huge amount of heat over long hours. Liquid cooling dessipates heat much faster, keeping CPU temperature lower so the system maintain top performance without lagging or thermal throttling. Additionly, it keeps the PC much quieter during live streaming, preventing loud, revving fan noise from being picked up by the microphone.